Marvin Gülhan

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Education

University of Bonn — Bonn, Germany
M.Sc. Computer Science · Focus: AI & Algorithms · Current grade: 1.1 (1.0 = best) — *Oct. 2025 – Mar. 2028*
Exchange semester: Universidad Politécnica de Madrid, Spain — *Feb. 2027 – Jun. 2027*

University of Bonn — Bonn, Germany
B.Sc. Computer Science · Focus: Algorithms · Graduated with honors · Grade: 1.2 (1.0 = best) — *Oct. 2022 – Sep. 2025*

- Thesis (Grade: 1.0): adapted Hybrid Genetic Search (PyVRP) to the Dynamic VRPTW and outperformed three state-of-the-art ACS baselines on the majority of dynamic Solomon instances

Research

Reinforcement Learning Amplifies Emergent Misalignment from Harmless Rewards — *EMNLP 2026*

- Showed that RL on rewards with no apparent harmful content still amplifies broad misalignment
- Implemented and evaluated mitigation techniques against RL-induced emergent misalignment

Multi-Agent Reinforcement Learning for Complex and Dynamic Graph-Based Planning Problems — *ICMCIS 2026*

- Developed graph reinforcement learning with GNNs and MAPPO for dynamic multi-agent airlift planning

Experience

Incoming AI & Operations Research Intern — Bonn, Germany
DHL Group · Deutsche Post AG — *Oct. 2026 – Dec. 2026*

- LLM agent-based query system over internal operational data and delivery-district optimization

Research Assistant — Bonn, Germany
Fraunhofer Institute FKIE — *Nov. 2023 – Present*

- Trained and evaluated GNN-based RL policies for dynamic multi-agent planning problems
- Developed online combinatorial optimization methods for dynamic routing and resource allocation under uncertainty
- Contracted Germany-wide OpenStreetMap road graphs with 10M+ edges into solver-ready instances

Competitions

Kaggle × Google DeepMind – Measuring Progress Toward AGI — *Apr. 2026*

- Co-authored PDPBench, a PDPTW benchmark evaluating frontier LLMs on request insertion, route construction, and full-solution generation

ICAPS 2024: AFRL Airlift Challenge – 2nd place — *Apr. 2024*

- Designed the core routing approach for heterogeneous aircraft on dynamic, disrupted networks
- Results published in Airlift Challenge: A Competition for Optimizing Cargo Delivery

Honors

Deutschlandstipendium (German National Scholarship) — *Oct. 2024 & Oct. 2025*

- Merit-based federal scholarship for outstanding academic achievement

Skills

Programming Languages: Python, C++
Libraries & Frameworks: PyTorch, Transformers, NumPy, Pandas, Gymnasium, NetworkX
Tools: Git, Linux, Docker, Slurm, OR-Tools
Languages: German (native), English (C1), Spanish (B1)